source → stages → sinks, once per frame

video → gray | median_blur | static_mask | mean_background | apply_mask →

- ffmpeg ← image
- display ← mask
- csv ← ctx.rows
- json ← ctx.metrics

Ctx — one per frame, handed down the chain

`image`	the working array — a stage rebinds it, never mutates it
`source`	the original BGR frame, untouched by anything
`index`	the frame number
`store`	side channel within one frame: \`mask\`, \`heat\`, \`boxes\`
`taps`	a named stage's output, kept for a sink to pick from
`metrics`	per-frame scalars — what the json sink writes
`rows`	per-object detail — what the csv and crops sinks write

Two kinds of state, deliberately apart: what survives the frame lives on the stage instance, what does not lives in \`store\`.

Why a side channel

A plain \`ndarray -> ndarray\` chain cannot express what the gas recipe needs:

\`mean_background\` accumulates the masked frame but subtracts from the unmasked one.

\`static_mask\` writes the mask, two later stages read it, and the frame carries on unchanged.

Nothing branches on runtime state; there is no state machine.